ASSIGNMENT-2

SUB DATE - 11/11/2022 (FRIDAY)

2. Explain how a GAN is trained.

GAN (Generative Adversarial Networks)

→ Generative Adversarial network or GAN, is a deep neural network framework which is able to learn from a set of training data and generate new data with the same characteristics of training data.

GAN Training

→ There are two aspects that make GAN more complex to train than a standard feed forward neural network:-

- The generator and the discriminator are really two neural networks which must be trained separately, but they also interact so they cannot be trained completely independently of each other.
- It is hard to identify exactly when a generative adversarial network has converged.
- Since the generator and discriminator have their own seperate loss functions, we have to train them separately.
- We can do this by alternating between the two:

→ we train the discriminator for one or more epochs, keeping the generator weights constant

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We repeat steps (1) and (2) until we determine that the network has converged

• Part 1:- The Discriminator is trained while the Generator is idle. In this phase, the network is only forward propagated and no back-propagation is done. The Discriminator is trained on real data for n epochs, and see if it can correctly predict them as real. Also, in this phase, the Discriminator is also trained on the fake generated data from the generator and see if it can correctly predict them as fake.

• Part 2:- The Generator is trained while the Discriminator is idle. After the discriminator is trained by the generated fake data of the Generator, we can get its predictions and use the results for training the Generator and get better from the previous state to fool and fool the Discriminator.

The above mentioned method is repeated for few epochs and then manually check the fake data if it seems genuine. If it seems acceptable, then the training is stopped, otherwise, its allowed to continue for few more epochs.

Ans
14/11/22